Lohith Kovuri

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Professional Summary

Data Analyst with 2+ years of Python and SQL experience in supply chain and payments, and an M.S. in Data Science. In personal projects, built AWS data pipelines in PySpark and Kafka, an LLM evaluation framework, and an agent that repairs data pipelines with human approval and an MCP tool server.

Skills

Programming: Python, SQL, R, FastAPI, REST APIs, Pydantic, pytest
Data Engineering: PySpark, Airflow, Kafka, Flink, dbt, Iceberg, AWS (S3, EMR, Glue), Terraform
Databases: Snowflake, Amazon Redshift, PostgreSQL, dimensional modeling, data validation
AI and ML: LLM APIs, tool calling, MCP server (Python SDK), LLM and RAG evaluation, XGBoost, SHAP, A/B testing
Visualization and Tools: Tableau, Power BI, Excel, Git, GitHub, Docker

Experience

Data Analyst | Uline | Remote, USA **Jul 2025 – Present**

- Automated about 50 daily shipping reports in Python, saving roughly **15 hours** of manual work a week.
- Wrote Python workflows that send SQL results to an LLM API to draft inventory and shipping reports, with checks.
- Added validation checks in Snowflake that cut reporting discrepancies by **22%** for the procurement team.
- Analyzed **75,000+** monthly inventory and order records in SQL to find bottlenecks and support demand planning.
- Built demand forecasting models in R that reduced forecast error by **15%** against baseline methods.
- Built Tableau and Power BI dashboards for 8 supply chain KPIs, cutting reporting turnaround by **30%**.

Data Analyst | Worldpay, LLC **Jan 2022 – Jul 2023**

- Helped consolidate merchant account data in AWS Redshift, cutting monthly reconciliation time by **30%**, and wrote Python scripts that pulled internal REST API data into gateway performance views.
- Wrote SQL fraud detection alerts and billing validation checks that cut billing discrepancies by **18%**.
- Automated payment gateway data cleaning with Python and Pandas, cutting manual prep time by **35%**.
- Built Excel and Power Query models for daily volume forecasts that fed treasury budget and cash planning.
- Defined data requirements with the product team in Jira and documented the data definitions in Confluence.

Personal Projects

PipelineMedic AI: Pipeline Repair Agent | github.com/LOHITHK22/PipelineMedic-AI

- Built a Python and FastAPI agent for a Kafka, Flink, PostgreSQL, and Airflow pipeline. Deterministic detectors flag schema drift, poison messages, consumer lag, and task failures before any LLM is involved.
- An LLM diagnoses the cause and proposes a typed repair plan. A risk policy runs low-risk repairs automatically and sends riskier ones to a human. Each repair is validated and rolled back if it fails.
- Exposed 17 typed tools through an MCP server (official Python SDK, stdio) with single-use approvals, audit logging, and 63 MCP tests.

RAG Sentinel: LLM Evaluation Framework | Python, XGBoost, SHAP

- Validated an LLM judge on 1,000 human-annotated responses (**0.82 Cohen's kappa**), weakest on multi-hop questions.
- Ran A/B tests on 4 prompt and embedding setups and found a **12%** relative faithfulness drop on multi-part questions.
- Built an XGBoost hallucination-risk classifier with SHAP (**0.89 ROC-AUC**) and a PSI and KS drift monitor.

AWS Streaming Data Platform | Kafka, PySpark, Iceberg, Glue, Airflow, dbt, Terraform

- Built a Kafka-to-S3/Iceberg pipeline across landing, raw, processed, and reporting layers, with a Kimball model for reporting in Athena and dbt.
- Wrote Glue PySpark jobs to validate records, drop duplicates, and load increments, with Airflow checks, error quarantine, and Terraform.

RetailBench Data Platform | PySpark, EMR, Airflow, Glue, Redshift

- Built a PySpark pipeline on EMR for **100+ GB** of TPC-H benchmark data, loading a dimensional Redshift warehouse.
- Cut job runtime from 24.6 to 13.8 minutes (**44%**) by tuning partitioning, broadcast joins, and skew handling.

Education

Rowan University | M.S. Data Science Sep 2023 – May 2025
Dr. MGR Educational and Research Institute | B.Tech Information Technology Sep 2019 – May 2023

Certifications

AWS Certified Data Engineer, Associate | AWS Cloud Practitioner | Microsoft Power BI Data Analyst Associate